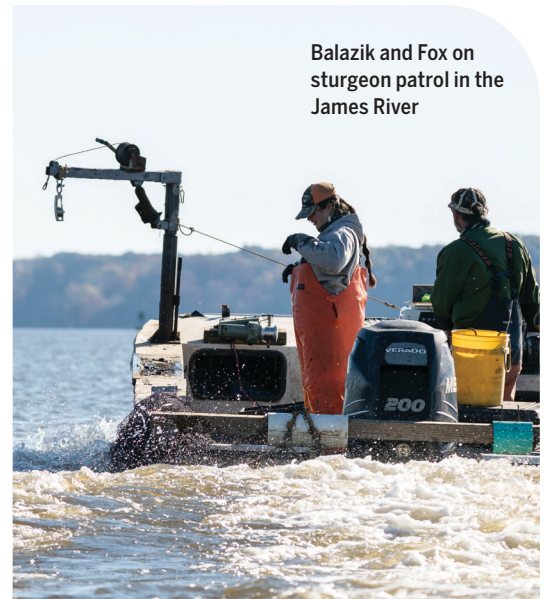




Dr. Matt Balazik, VCU faculty researcher (R), and Carrie Fox, fisheries technician for the Upper Mattaponi Tribe, with juvenile Atlantic sturgeons in the James River



Balazik and Fox on sturgeon patrol in the James River



Measuring a juvenile Atlantic sturgeon in the James River near Hopewell

Ancient Swimmers

Atlantic sturgeon survived the dinosaurs, a mass extinction, and centuries of overfishing. Now, a dedicated team of James River scientists is making sure they survive us.

By JOHN PAGE WILLIAMS | Photography by LYNDIA RICHARDSON

ON A JAMES RIVER BOAT TRIP IN 2004, a science class of middle schoolers spotted a little 6-inch fish wriggling in the vessel's trawl net. To the students, it may have been just any old catch. But to the boat's captain and crew, it was cause for genuine excitement: a young Atlantic sturgeon. The juvenile had likely spawned the previous September—offering the first evidence in decades that this endangered mega-species was reproducing naturally somewhere in the Chesapeake.

Known as “fossil fish,” sturgeons can be traced back nearly 200 million years, all the way to the Early Jurassic period. Sizing up to 800 pounds and 13 feet long, they typically live for nearly 60 years.

Virginia Commonwealth University's Rice Rivers Center in Charles City County has been a hub for Virginia's Atlantic sturgeon research since 2007, and the scientist at the center of it all is Dr. Matt Balazik, a faculty researcher at VCU's School of Life Sciences and Sustainability.

Balazik grew up prowling the river around Hopewell, captivated by stories of the river's “mythical beast”—the sturgeon. Comfortable around boats and strong enough to handle the big fish, he signed on for graduate study at the Rice Center as the research developed. While overall Atlantic sturgeon numbers remain low, the population has shown encouraging signs of improvement, thanks largely to the 40-year coastwide morato-

rium on harvesting enacted in 1998. Recovery, however, is a slow-motion affair: Sturgeon don't mature until well into their teens, and females typically spawn only every other year—meaning even with solid protections in place, it takes decades for depleted stocks to meaningfully rebound.

Despite a tremendous amount of data collected since 1998, the scientific community still considers the Atlantic sturgeon a “data-poor” species—a humbling reminder of how much remains unknown about one of the river's most ancient inhabitants. “Don't think everything about them is figured out,” Balazik says. “We need multiple types of information that improve the abilities of fishery agencies to manage the stock.” The gaps he's working to fill include mark-recapture studies to track growth rates and population trends, monitoring the movements of individually tagged fish outside spawning seasons, genetic research to understand how fish from different East Coast river stocks relate to one another, and investigating how human alterations to the river affect the sturgeon's lifecycle. These fish depend on healthy bottom-dwelling communities—worms, shellfish, small finfish—to feed upon and thrive.

Mark-recapture studies reflect the deep expertise that Balazik and his team have built over 20 years of netting for sturgeon in the James and beyond. Careful tagging and handling techniques keep the fish comfortable while aboard and allow

them to swim away unharmed. The program has tagged more than 2,350 individual fish with PITs (Passive Integrated Transponders) that identify them upon recapture. Around 400 of those fish have also been implanted with acoustic transmitters that “talk” to a network of underwater receivers via unique identifier codes—an elegant technological conversation happening silently beneath the river’s surface.

Tagging technology is evolving quickly, and over time, the accumulated data is painting a

“Our goal is to do the science that will lead to better environmental policies and practices that protect species like the sturgeon.”

— Greg Garman, Director, Rice Rivers Center

vivid portrait of these long-lived coastal wanderers. Immature Atlantic sturgeon—those fewer than 12 years old—travel widely up and down the Chesapeake, along the entire East Coast, and as far as Iceland when at sea. Data from radio-tagged, fall-run James River males reveals that in just over a month, they can log several hundred miles in multiple trips upriver and back, restlessly seeking spawning females.

“Some fish seem to develop migration patterns,” Balazik observes—and nowhere is that more evident than in the travel records of two tagged fish his team has named “Stuart” and “Chuck.” Stuart’s tags reveal a summer of wandering north to Philadelphia and Long Island in search of good meals, before he reliably returns to the James by early September, soon ping- ing receivers around the spawning reefs above the Presqu’ile National Wildlife Refuge, across from Hopewell. Chuck, meanwhile, follows his own dependable rhythm: turning up faithfully in the Chesapeake each March, lingering in the lower James through summer, running upriver to spawn in the fall, and disappearing again by November—only to spend each winter ping- ing the receivers of Balazik’s fellow sturgeon researchers along the coast of South Carolina. These fish, it turns out, have very definite preferences.

Genetic studies from fin clips reveal that the James River sturgeon stock is distinctively its own, with subtle differences even between the river’s spring and fall fish. Among other Chesa-



Known as “fossil fish,” sturgeons can be traced back to the Early Jurassic period. They can weigh up to 800 pounds and exceed 13 feet in length.

apeake river stocks, the fish of the Eastern Shore’s Nanticoke River—including those in its major tributary, Marshyhope Creek—are genetically related to York River fish, which spawn in the Pamunkey and Mattaponi tributaries. “We’re trying to figure out the nuances because every river is different,” Balazik says.

The threats, unfortunately, are real and persistent. Ship strikes pose a serious danger: Sturgeon hug the river bottom while feeding, but during migration they rise to depths that put them squarely in the path of ship hulls and propellers. “We’re trying to pinpoint the areas of greatest risk as we work out migration tracks and timing,” Balazik says. Troubling spawning seasons yielding fewer fish add to the concern.

Rice Rivers Center Director Greg Garman frames the mission with clarity: “Our goal is to do the science that will lead to better environmental policies and practices that protect species like the sturgeon.” Knowing where these fish are, when they arrive, and what is putting them at risk is the essential foundation for turning the tide—and hopefully driving better recruitment numbers and survival rates in the years ahead.

“There’s still so much out there to be discovered,” says Balazik—though he’s candid that two decades of sturgeon fieldwork in all seasons have taken a real physical toll. One of his priorities now is fostering the next generation of researchers to carry the program forward. “Minimize threats, build a better path to restoration,”

he says. “Fieldwork—it’s all luck. You just need to improve your chances of getting lucky. Stop monitoring the problem and start fixing it.”

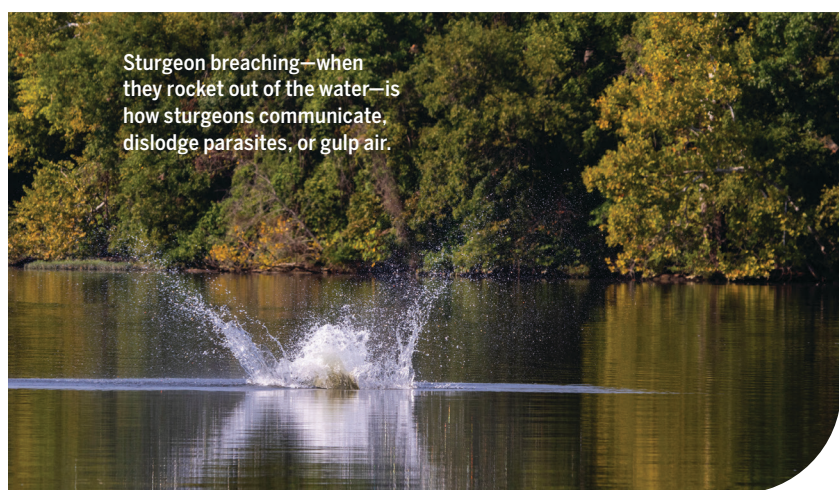
Fisherman, writer, educator, and skipper of First Light, John Page Williams is a retired Chesapeake Bay Foundation naturalist and an editor-at-large for Chesapeake Bay Magazine.

THE GREAT RETURN

Atlantic sturgeon are breaching on the James again. Here’s how to witness—and toast—the comeback.

While the research continues, the presence of Atlantic sturgeon in the James is now understood enough that the public is curious. In September, the James River Association will offer public tours launching from Jordan Point Marina in Hopewell during James River Week—Sept. 12–19—where passengers may spot 5–6-foot-long male fish breaching (leaping) clear from the river’s surface. Registration opens in August on JRA’s website. TheJamesRiver.org

To celebrate the season and its potential for a new generation of young sturgeon, as well as a tribute to decades of hard work by conservationists to restore the James River as a bounty of vibrant, aquatic life, Richmond’s Hardywood Brewery will offer The Great Return, a pale ale whose sales support JRA’s mission to be guardians of the James. Hardywood.com



Sturgeon breaching—when they rocket out of the water—is how sturgeons communicate, dislodge parasites, or gulp air.